

Complexity and Label-Budget Trade-offs in Probability Updating and Class-Conditional Conformal Prediction under Natural Tabular Shifts

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Abstract

After transfer across populations, regions, or time, a predictive model can retain discrimination while its probability estimates deteriorate—a failure mode that discrimination metrics alone cannot reveal. We asked whether limited target labels acquired by label-blind random sampling could reliably support probability updating and class-conditional conformal prediction under natural tabular shifts. Across five natural-shift tasks, two base models, four probability-updating methods, and absolute budgets of 25–500 labels, each target-fitted update was compared with its same-method source-fitted baseline; non-estimable attempts were retained in feasibility accounting. Increasing the budget improved feasibility where it was initially limited and reduced observed harm, but label count alone did not ensure benefit across tasks, models, and calibrators. The task- and model-equal mean log-loss benefit for intercept-only updating increased from -0.0001 at 25 labels to $+0.0088$ at 500, while its material-harm rate among successful finite pairs fell from 43.1% to 1.0%. Sigmoid's mean benefit changed from -0.0255 to $+0.0085$, and its corresponding harm rate fell from 74.0% to 1.7%. Isotonic remained harmful on average throughout the tested grid: mean

benefit changed from -0.4218 to -0.0172 , while its corresponding harm rate fell from 99.7% to 73.2%. No method at any tested absolute budget met the prespecified $+0.002$ log-loss improvement criterion in all 10 task-model cells. Among jointly successful Standard-Mondrian pairs, the within-cell median Mondrian-minus-Standard difference in observed worst-class coverage was positive in all 50 task-model-budget cells, but mean prediction-set size also increased, and only 30.1% of available Mondrian attempts yielded finite thresholds at 25 labels. In the finite-threshold sensitivity analysis at 25 labels (8/10 task-model cells), the equal-cell mean worst-class-coverage difference was $+0.0275$ rather than $+0.1285$, and the mean set-size difference was $+0.0013$ rather than $+0.3280$. These findings argue against a universal label-count rule: within the evaluated grid, reliability was specific to task, model, method, class support, and budget.

Keywords: probability calibration; model updating; distribution shift; tabular data; conformal prediction; class-conditional coverage; label budget; reliability

1. Introduction

Predictive models deployed across populations, regions, or time periods may retain useful ranking performance while their probability estimates cease to reflect target-domain outcomes. Reliable probabilities matter whenever predictions inform thresholds, resource allocation, or risk communication, so a modest set of newly observed target labels is often used to update a model without rebuilding it. Clinical and broader shift studies have documented calibration drift, heterogeneous changes across models, and the need to assess probability quality separately from discrimination (Davis et al. 2017; Levy et al. 2022; Ovadia et al. 2019; Van Calster et al. 2019). The practical question is therefore not only whether a source model transfers, but whether limited labeled target data can improve its reliability.

Such updating is not automatically beneficial. A method may be unavailable when a blind sample contains too few observations from one class, and an estimable update may still increase log loss or Brier score through variance or overfitting. Earlier work already shows that recalibration behavior depends on sample size and method complexity: parsimonious prediction-model updates can be preferable with scarce data, while flexible procedures need more support and can harm an already well-calibrated predictor (Steyerberg et al. 2004; Niculescu-Mizil and Caruana 2005; Ojeda et al. 2023). Class-conditional conformal procedures face a related feasibility constraint because finite classwise thresholds require sufficient calibration support in each class (Ding et al. 2023; Liu et al. 2026).

Label budget, target prevalence, base model, and update complexity consequently need to be considered together. The same total budget can contain very different numbers of minority-class labels across targets, and the same number of labels can represent very different fractions of the available target

population. Moreover, the data needed by an intercept correction differ from those needed by sigmoid or isotonic calibration, just as the data needed for a marginal conformal threshold differ from those needed for separate classwise thresholds. Finite-sample recalibration and model-updating studies have established these dependencies in synthetic, clinical, or independently calibrated settings (Sun et al. 2023; Vergouwe et al. 2017; Davis et al. 2020). What remains operationally important is their joint behavior under a fixed natural-shift protocol.

Existing research addresses the constituent questions from several directions. Probability-calibration studies compare parametric and nonparametric updates; model-maintenance studies examine recalibration under temporal or external validation; tabular-shift benchmarks characterize heterogeneous geographic, temporal, and population shifts; and conformal work studies marginal, classwise, and sparse-class coverage (Ojeda et al. 2023; Gardner et al. 2023; Liu et al. 2023; Ding et al. 2023). Other work adapts calibration or uncertainty under explicit covariate- or label-shift assumptions, sometimes without labeled target outcomes (Park et al. 2020; Popordanoska et al. 2024; Podkopaev and Ramdas 2021).

This study addresses a joint empirical question: how benefit, harm, estimability, budget heterogeneity, and prediction-set cost interact under a common label-blind protocol for natural tabular domain shifts. We evaluate public binary tabular tasks with one record per entity, acquire target labels by label-blind random sampling, and compare updates against same-method budget-zero source baselines. This design keeps failed or unavailable cells visible and connects probability adaptation with conformal feasibility and efficiency. The intended contribution is empirical and protocol-based rather than methodological: a bounded quantitative map for the evaluated setting, not a general ranking of calibrators or a deployment rule.

One primary contribution concerns complexity, budget, benefit, and harm. Across the primary five-task set, logistic regression and XGBoost base models, and label-blind random target-label budgets from 25 to 500, the mean benefit of intercept updating became positive at a lower tested budget than sigmoid recalibration. Isotonic calibration retained high observed material-harm rates among successful finite pairs throughout the prespecified test grid. This result describes the observed ordering and harm frequencies within these tasks, models, methods, and budgets; it does not establish a general calibrator ranking.

The second primary contribution concerns budget heterogeneity. Within the same five-task, two-model, four-calibrator grid, no tested absolute target-label budget produced a median log-loss benefit of at least 0.002 in every task-model-calibrator cell. The fractions of target adaptation pools represented by a given absolute budget varied substantially; the design did not test a common relative-budget grid and therefore does not identify a universal fraction threshold. Leave-one-task-out and leave-one-source-family-out analyses assess how strongly this

conclusion depends on individual tasks or data families, while a separate small-domain stress case illustrates why nominal label counts cannot be interpreted without the size and class composition of the target pool.

The secondary contribution concerns class-conditional reliability. Among jointly successful paired comparisons over the primary five-task set, two models, and label-blind random budgets from 25 to 500, the median Mondrian-minus-standard difference in worst-class coverage was positive in every evaluated task-model-budget cell. The increase was accompanied by budget-dependent finite-threshold availability and larger prediction sets, including more binary doubletons at smaller budgets. We therefore report conditional metric differences, estimability, finite-threshold rates, and set cost together. Auxiliary analyses document eligibility boundaries and show that small AUROC changes did not rule out worse proper scores, but neither auxiliary result is treated as a central contribution.

2. Related Work

2.1 Probability calibration and model updating

Post-hoc probability calibration includes parametric corrections such as logistic or sigmoid scaling and more flexible procedures such as isotonic regression. Comparative work has long shown that performance depends on both the underlying learner and the amount of calibration data. Niculescu-Mizil and Caruana found sample-dependent ordering between Platt scaling and isotonic regression and observed that recalibration can harm models that are already well calibrated (Niculescu-Mizil and Caruana 2005). Broader comparisons likewise show heterogeneity across probability models, structural differences, prevalence, and calibration sample size; regression-based approaches often behave more stably when data are limited, whereas isotonic behavior is especially sample-sensitive (Ojeda et al. 2023). Finite-sample analyses further connect calibrator complexity with risk and show how the useful resolution of a recalibration method should change with available data (Sun et al. 2023). These studies directly motivate, and constrain the interpretation of, our complexity-budget analysis.

Prediction-model updating research provides an equally direct precedent. Clinical external-validation studies compare retaining a model with adjusting its intercept, revising its calibration slope, or changing more coefficients. Their results favor matching update complexity to target sample size and using shrinkage or staged selection to control overfitting (Steyerberg et al. 2004; Vergouwe et al. 2017). Longitudinal maintenance studies also show that the timing and extent of updating can differ by algorithm, so a single updating policy need not serve every model (Davis et al. 2020; Levy et al. 2022). Recent work under natural image shifts similarly reports heterogeneous trade-offs between simple post-hoc and shift-specific calibration procedures (Roschewitz et al. 2025). Our study does not propose a calibration operator. Its narrower increment is a paired

benefit-and-harm audit of prespecified calibrators under blind target-label budgets, with unavailable fits retained rather than silently removed.

2.2 Natural tabular distribution shift

Natural tabular shifts differ from random train–test splits because domain membership can represent geography, calendar time, institution, or population group. TableShift organizes public binary tabular tasks around such naturally occurring domains and documents substantial heterogeneity in out-of-domain performance (Gardner et al. 2023). WhyShift complements benchmark-style evaluation with a language for describing shift structure and shows that interventions can behave differently across datasets and shift types (Liu et al. 2023). These works establish both the natural tabular setting and the importance of dataset-specific interpretation. The present study uses that setting for a different endpoint: recovery of probability and prediction-set reliability with a limited, label-blind target sample.

Reliability under shift has also been studied under more specific assumptions or in other modalities. Covariate-shift methods use importance weighting or domain adaptation to transfer calibration without labeled target outcomes (Park et al. 2020), while label-shift methods estimate or correct target calibration under assumptions about changing class proportions (Popordanoska et al. 2024; Podkopaev and Ramdas 2021). Medical studies show that natural temporal or population shifts can affect discrimination and calibration differently and that reliability updates may persist across clinically relevant subpopulations (Davis et al. 2017; Guo et al. 2023; Kapash et al. 2025). These approaches answer important but distinct questions. Our protocol does not assume that the observed natural shifts are pure covariate or label shift; it instead measures what happens when a prespecified number of target outcomes is sampled without observing labels during acquisition.

2.3 Conformal prediction and class-conditional coverage

Conformal prediction converts model scores and a labeled calibration sample into prediction sets with finite-sample coverage properties under its stated sampling conditions (Angelopoulos and Bates 2023). Adaptive scores can improve set efficiency (Romano et al. 2020), and weighting or sequential adaptation can address particular forms of covariate shift or nonexchangeability (Tibshirani et al. 2019; Foygel Barber et al. 2023; Gibbs and Candès 2024). Few-shot conformal work also shows that very small calibration samples can yield uninformative sets and studies how auxiliary tasks may improve efficiency (Fisch et al. 2021). Our setting instead uses no auxiliary-task transfer and treats the finite or infinite status of each threshold as an observed protocol outcome.

Marginal coverage can conceal weak performance for infrequent classes, motivating class-conditional or Mondrian procedures. Prior work establishes that rare classes may lack enough calibration observations for finite classwise quan-

tiles and that better classwise coverage can require larger prediction sets (Ding et al. 2023; Liu et al. 2026). Recent methods seek different points on this coverage–efficiency trade-off through augmented ranks, prevalence adjustment, interpolation, or hierarchical fallback (Shi et al. 2024; Ding et al. 2026; Papaioannou et al. 2026). Binary anomaly-detection evidence likewise shows that class-conditional coverage improvements can be accompanied by overcoverage or efficiency costs under extreme imbalance (Althani 2026). Accordingly, our Standard–Mondrian comparison is a budget-response audit of an established trade-off, not an overall method ranking.

2.4 Positioning of the present study

Prior work has examined complexity-sensitive recalibration, algorithm-dependent model updating, natural tabular shifts, calibration under explicit shift assumptions, sparse-class conformal thresholds, coverage–set-size trade-offs, and the distinction between discrimination and probability reliability. This study integrates these components in one prespecified empirical protocol that jointly reports benefit, harm, estimability, target-pool fraction, and prediction-set cost across the primary five-task set, two base models, target-label budgets of 25–500, and four probability updates. The contribution is empirical and protocol-based rather than methodological or theoretical.

No complete duplicate of that full protocol was identified in the bounded literature audit, but this observation is limited to the searched and verified record and is not evidence of global absence. The closest studies overlap substantially with individual questions while differing in data modality, shift assumptions, sample regime, adaptation information, or reported outcomes. We therefore organize the manuscript around scoped quantitative boundaries: when evaluated calibrators became beneficial or harmful, whether one budget satisfied the prespecified criteria across the tested grid, and how class-conditional coverage changed with threshold feasibility and set size. All conclusions remain conditional on the selected tasks, models, methods, acquisition rule, and budgets.

3. Study Design and Scope

We studied a constrained target-adaptation setting in which a probabilistic binary classifier had already been trained in a source domain and a limited number of labels could subsequently be acquired from a shifted target domain. The predictor itself remained frozen. The decision under study was therefore not whether to retrain the predictive model, but whether a particular probability or conformal update was usable and beneficial at a given target-label budget. Label acquisition was uniform and label-blind: target outcomes were revealed only after the sampled entities had been selected. This design separates adaptation from active learning, oracle stratification, and outcome-aware monitoring.

The empirical scope comprised six public binary tabular tasks from five source

families. Five tasks formed the common analysis set because every prespecified budget from 25 to 500 labels was available. College Scorecard was retained as a separate small-domain stress case because its target adaptation capacity was 107 entities; budgets 250 and 500 were recorded as unavailable rather than imputed, refilled, or extrapolated. The primary evidence concerned probability-update complexity and budget (H2) and the absence of a common tested absolute budget across tasks and methods (H3). The Standard–Mondrian class-conditional coverage trade-off (H5) was secondary. Class-support feasibility (H1), source-to-target metric changes (H4), and simulations were auxiliary. H6 was designated `not_in_final_scope`; no sampling-policy comparison was conducted.

Repetition represented three distinct sources of variation. Entity partitions were generated for split seeds 0, 1, and 2. Logistic regression used model seed 0, whereas CPU XGBoost used model seeds 0 and 1. Within each resulting task–split–model shard, positive-budget acquisition used 100 adaptation seeds. They were numbered 0–99, derived from the configured count. The budget-zero source reference used adaptation seed -1 . This produced 54 prespecified model configurations: six tasks, three split seeds, and three model/seed combinations per split. Split, model, and adaptation seeds were stored separately and were not treated as interchangeable replicates.

The study design, archived experiment results, independent scientific analysis, and allowed claims were fixed before manuscript preparation; their version identifiers and hashes are reported in the supplement and repository audit. Manuscript figures and tables use only these archived results. Subsequent reproduction checks were executed separately and did not replace or modify the outputs used for the reported analyses.

The unit of a probability decision was a task–base-model–calibrator–budget combination supported by paired runs, rather than an isolated run or a pooled collection of rows. The unit of the conformal decision was the same task and base model at a given budget, with Standard and Mondrian evaluated on an identical target sample and score source. This distinction matters because a large task or a model with two random seeds would otherwise contribute more rows and dominate a pooled mean. The design instead asks whether the direction and risk profile remain visible after collapsing repeated acquisitions within tasks and then weighting scientific units equally. It also treats usability as part of the decision: a candidate update is described by availability, eligibility, conditional performance, and, for Mondrian, finite-threshold status. These are complementary coordinates rather than a sequence in which missing cells are discarded before accuracy is summarized.

Study design and the seven mutually exclusive data roles. Target adaptation and final evaluation remain entity-disjoint; all methods within a run share a label-blind random nested target sample.

4. Data and Natural Domain Shifts

The primary five-task set contained two 2018 American Community Survey (ACS) 1-year Public Use Microdata Sample tasks and one task each from the Behavioral Risk Factor Surveillance System (BRFSS), the National Health and Nutrition Examination Survey (NHANES), and the UCI diabetes-readmission data (U.S. Census Bureau 2018, 2019; Clore et al. 2014). ACS Income used the Folktables adult filter (age above 16 years, personal income above \$100, positive usual weekly hours, and person weight at least one), labeled personal income PINCP at least \$56,000 as positive, and contrasted New England (DIVISION = 1, target) with other census divisions (source). This threshold is a project adaptation of the benchmark rather than an exact reproduction of the strict $\text{PINCP} > 56,000$ TableShift task (Ding et al. 2021; Gardner et al. 2023). ACS Food Stamps selected the household reference-person row (RELPH = 0), ages 18–61, personal—not household—income PINCP at most \$30,000, and HUPAC codes 1–4, then predicted household food-stamp receipt (FS = 1). Those HUPAC codes include its no-children category; the cohort must not be described as restricted to households with children. East South Central (DIVISION = 6) was the target and all other divisions were the source.

BRFSS combined the 2015, 2017, 2019, and 2021 public-use files (Centers for Disease Control and Prevention 2015, 2017, 2019, 2021). DIABETE3 (2015/2017) or DIABETE4 (2019/2021) code 1 was positive; codes 2 (pregnancy only), 3 (no), and 4 (prediabetes/borderline) were negative, and refused, unknown, and missing responses were excluded. _PRACE1 = 1 (White race) defined the source; codes 2–6 (Black or African American, American Indian or Alaska Native, Asian, Native Hawaiian or other Pacific Islander, or other race) defined the target. This is a race-code partition, not a White-non-Hispanic contrast, and _PRACE1 and the diabetes item were excluded from predictors. NHANES merged cycle-qualified respondent identifiers across cycles from 1999–2000 through 2017–2018 and retained complete public-use LBXPBP, INDFMPIR, and RIDAGEYR records at age 18 or younger (National Center for Health Statistics 2020a, 2020b, 2020c). The target had family poverty-income ratio at most 1.3, the source had a ratio above 1.3, and the positive label was blood lead at least $3.5 \mu\text{g/dL}$. The $3.5 \mu\text{g/dL}$ value was used as a benchmark cutpoint informed by the CDC blood lead reference value, not as a toxicity or clinical-diagnosis threshold; the age range also extends beyond the 1–5-year population used to derive that reference value (Ruckart et al. 2021). Public-use lead measurement availability was not uniform over cycles, and no survey weighting was used.

In the UCI task, admission_source_id = 7 (emergency room) defined the target and all other retained admission sources defined the source. The positive label was readmitted != NO, so both <30 and >30 categories were positive; it is not the original study's narrower 30-day endpoint (Clore et al. 2014; Strack et al. 2014). A deterministic stable hash selected one patient index encounter, which is not a chronological first-encounter rule, and the prediction time was hospital discharge. College Scorecard used the fixed MERGED2018_19_PP.csv release,

rather than a generic calendar-2018 snapshot (U.S. Department of Education 2026b, 2026a, 2025). Public and private nonprofit institutions (`CONTROL` = 1 or 2) formed the source, private for-profit institutions (`CONTROL` = 3) formed the target, and `C150_4` $\leq$ 0.50 denoted completion at or below 50% among the applicable IPEDS first-time, full-time cohort within 150% of normal program time. It is not a pooled all-student completion measure. College contributed only the lower-budget stress analysis. Across all tasks, domain indicators, direct identifiers, label variables, and audit keys were excluded from predictors. These observational partitions do not identify causal effects of geography, race, poverty, admission source, or institutional control.

Every task obeyed a one-entity-one-row rule. The entities were an ACS person, an ACS household represented by its reference-person row, a BRFSS respondent, an NHANES respondent identified within survey cycle, a UCI patient represented by the stable-hash index encounter, and a College Scorecard institution. Audits found no duplicate entity in the final task tables. Entity identity, rather than a potentially repeated record, was the leakage-control unit.

For each split seed, entities were assigned once to seven mutually exclusive roles: `source_train`, `source_tune`, `source_probability_calibration`, `source_conformal_calibration`, `source_id_test`, `target_adaptation_pool`, and `target_final_test`. A seed-specific hash allocated source entities approximately 60%/10%/10%/10%/10% to the five source roles and target entities 40%/60% to adaptation and final testing, after which prespecified label-blind caps were applied. Maximum source-role sizes were 100,000 for training, 20,000 each for tuning and the two calibration roles, and 30,000 for source in-domain testing; target final testing was capped at 50,000, while the target adaptation pool was uncapped. Smaller roles retained all available entities. Model selection used only source training and tuning. The two source calibration roles were disjoint, and target-final labels were used only for evaluation.

Target acquisition used `blind_random` only. For each dataset, split seed, and adaptation seed, entities in the target adaptation pool were ordered by a deterministic hash that did not accept labels as input. Budgets 25, 50, 100, 250, and 500 were nested prefixes of this ordering, so all methods evaluated at a given run shared the same selected entities and `sample_hash`. A budget exceeding the pool was marked `not_available`; there was no resampling after outcomes were observed. Results were reported on both the absolute budget axis and the fraction of the available target adaptation pool because identical label counts represented different sampling intensities across tasks.

All task registries were finalized before the writing stage and linked to processed-data checksums. Table 1 reports positive-outcome rates in the audited full source and target domains, whereas the adaptation experiments used seed-specific entity-disjoint pools derived from those domains. The two quantities should not be conflated: the full-domain rate describes the observed shift, while the pool composition determines the realized support probabilities for a particular split. Public availability also does not make the sources interchangeable.

ACS and BRFSS are large surveys, NHANES combines survey and laboratory measurements, the UCI table originates from hospital encounters after patient-level indexing, and College Scorecard is an institution-level snapshot. Keeping source family and entity unit explicit prevents row counts from being interpreted as a single homogeneous sampling frame.

Table 1. Task definitions, natural domains, entity units, audited counts, and analysis roles.

Task	Source domain	Target domain	Outcome / inclusion	Entity	Period / version	Source n (positive %)	Target n (positive %)	Analysis role
ACS Income	DIVISION != 1 (outside New England)	DIVISION = 1 (New England)	PINCP $\geq$ \$56,000 after Folktables adult filter	person	2018 ACS 1-year PUMS	1,575,270 (32.9%)	84,346 (40.3%)	common five
ACS Food Stamps	DIVISION != 6 (outside East South Central)	DIVISION = 6 (East South Central)	FS = 1; reference-person row aged 18–61, PINCP $\leq$ \$30,000, HUPAC 1–4	household (RELP = 0)	2018 ACS 1-year PUMS	276,050 (23.0%)	21,932 (27.1%)	common five
BRFSS Diabetes	_PRACE1 = 1 (White)	_PRACE1 = 2–6 (Black/AIAN/Asian/NHPI/Other)	DIABETE3/4 = 1; codes 2–4 negative	respondent	BRFSS 2015, 2017, 2019, 2021	1,409,220 (12.6%)	287,074 (17.0%)	common five
NHANES Lead	INDFMPIR > 1.3	INDFMPIR $\leq$ 1.3	LBXBPB $\geq$ 3.5 μ g/dL among age $\leq$ 18	SEQN within cycle	NHANES 1999–2000 to 2017–2018	14,759 (2.6%)	12,740 (8.0%)	common five; rare-positive
Diabetes Readmission	admission__source_id != 7	admission__source_id = 7 (emergency room)	readmitted != NO (both <30 and >30 positive)	patient index encounter	UCI encounters, 1999–2008	31,833 (30.7%)	37,835 (35.7%)	common five
College Scorecard	CONTROL = 1 or 2 (public/nonprofit)	CONTROL = 3 (private for-profit)	C150_4 $\leq$ 0.50	institution (UNITID)	MERGED2018_19_PP.csv	1,986 (46.7%)	268 (66.4%)	separate small-domain stress

5. Methods

We evaluated logistic regression and CPU XGBoost as two contrasting base predictors. Numeric features were median-imputed and standardized. Categorical features were one-hot encoded with unknown-category handling and a minimum category frequency of five. For each split and model seed, two prespecified hyperparameter candidates were fitted on `source_train`; the candidate with higher AUROC on `source_tune` was retained. Logistic regression used the liblinear solver, no class weighting, a maximum of 1,000 iterations, and $C \in \{0.1, 1.0\}$. XGBoost used 120 depth-four trees, learning rate 0.05, row and column subsampling 0.9, histogram tree construction on CPU, four CPU threads, and log loss as the evaluation metric. Its two candidates paired minimum child weight 1 with L_2 regularization 1.0 and minimum child weight 5 with L_2 regularization 2.0. Once selected, the entire preprocessing–prediction pipeline remained fixed.

Probability updates acted on the base-model probability p . The uncalibrated method was the identity mapping and was stored at budget zero only. The intercept-only maximum-likelihood update added a fitted scalar offset to $\text{logit}(p)$, retaining unit slope. The Jeffreys-smoothed intercept update used the same unit-slope form but matched the mean fitted probability to $(n_1 + 0.5)/(n + 1)$, permitting a finite fit even when the sampled target labels belonged to one class. Sigmoid calibration fitted an unpenalized logistic regression with free intercept and slope to the base-model log odds. Isotonic calibration fitted a bounded monotone mapping from the base probability to the binary outcome. Base probabilities were clipped to $[10^{-6}, 1 - 10^{-6}]$ only where logit or log-loss numerical stability required it; ranking metrics retained the unclipped ordering. Sigmoid calibration used scikit-learn logistic regression with $C = \infty$, the `lbfgs` solver, and at most 1,000 iterations. Both intercept solvers used SciPy's Brent root finder on $[-40, 40]$. Isotonic regression used `out_of_bounds="clip"`, `y_min=0`, and `y_max=1`.

Method eligibility was explicit. Intercept-only maximum likelihood and sigmoid required at least one observed label from each class. Isotonic required at least 25 labels in total, at least five labels per class, and at least 10 unique input scores. Jeffreys smoothing required a nonempty sample but did not require both classes. At budget zero, each fitted method was trained on `source_probability_calibration`. At a positive budget, it was instead fitted to the label-blind random target prefix and evaluated on `target_final_test`. Every adapted probability method was compared with its own source-fitted budget-zero version; a shared uncalibrated comparator was not substituted for these same-method baselines.

We evaluated Standard split conformal and outcome-class-conditional Mondrian conformal prediction at nominal coverage 0.90. For candidate class y , nonconformity was $1 - p_y$. Standard conformal used one threshold from all calibration scores, whereas Mondrian used separate thresholds for classes 0 and 1 and applied the threshold corresponding to each candidate label. The finite-sample

order was $\lceil (n+1)(1-\alpha) \rceil$. Here $\alpha = 0.10$; a requested order above the available calibration size produced an infinite threshold rather than a silently altered quantile. Mondrian required at least five calibration labels per class, although estimability did not itself guarantee that both classwise thresholds were finite. Two fixed score sources were evaluated: base uncalibrated probabilities and probabilities transformed by a sigmoid fit on the disjoint source probability-calibration role.

Budget-zero conformal thresholds were fitted on `source_conformal_calibration`. Positive-budget thresholds were refreshed from the same label-blind random target prefix used by the probability methods, preserving exact sample pairing. A prediction set included every class whose candidate-label nonconformity did not exceed its threshold. For binary outcomes, set sizes therefore ranged from zero to two; a doubleton was the full two-label set. Standard and Mondrian results were always paired on dataset, split, model, model seed, adaptation seed, budget, score source, evaluation split, and sample hash.

Every attempted cell retained its computational state. `success` denoted a completed fit and evaluation. `not_available` denoted a prespecified budget exceeding the target pool. `not_estimable` denoted failure of a declared support or method-eligibility rule. `failed` was reserved for a technical exception. Separate indicators recorded whether a method was estimable and whether its conformal threshold was finite. These categories were never collapsed into a single missing value, and unavailable or non-estimable attempts were not counted as neutral performance outcomes.

The methods form a complexity comparison, not a claim that their parameter counts alone determine risk. Intercept-only MLE estimates one target offset; the smoothed version modifies its small-sample target; sigmoid estimates an intercept and slope; and isotonic can adapt a monotone curve to the observed scores. Their behavior can also depend on source score geometry, target prevalence, and the base predictor. For this reason, each adapted row retained the observed positive and negative counts, number of unique input scores, target-pool fraction, and exact sample hash. Conformal score sources were likewise fixed rather than selected after target performance was observed. These records allow complexity to be discussed alongside its actual support and sampling context without reducing the experiment to a single nominal degrees-of-freedom ordering.

Table 2. Evaluated methods, support rules, and reported outcomes.

Method	Family	Frozen support rule	Comparison role	Reported outputs
Uncalibrated	Probability reference	budget 0 only	no target fit	AUROC, AUPRC, Brier, log loss, calibration intercept/slope
Intercept-only MLE	Probability update	≥ 1 label per class	same-method source baseline	Brier, log loss, calibration intercept/slope
Jeffreys intercept	Probability update	nonempty sample; smoothed	low-budget availability comparator	Brier, log loss, calibration intercept/slope
Sigmoid	Probability update	≥ 1 label per class	same-method source baseline	Brier, log loss, calibration intercept/slope
Isotonic	Probability update	$n \geq 25$; ≥ 5 /class; ≥ 10 unique scores	same-method source baseline	Brier, log loss, calibration intercept/slope
Standard	Conformal	total-size quantile	nominal 0.90 marginal sets	marginal/worst-class coverage, mean set size, doubletons
Mondrian	Conformal	≥ 5 /class to estimate; exact finite-quantile check	nominal 0.90 class-conditional sets	marginal/worst-class coverage, mean set size, doubletons, finite threshold

Ethics Statement

This study involved only secondary analysis of publicly available, de-identified datasets. The author did not recruit or interact with participants, collect new human-participant data, conduct interventions, or access direct personal identifiers. On this basis, separate institutional ethics review was not required, and no approval or determination number was issued.

6. Evaluation Protocol

Probability evaluation included AUROC, AUPRC, Brier score, binary log loss, calibration intercept, calibration slope, and absolute deviations of the two calibration coefficients from their ideal values zero and one. Skill scores used a prevalence-only predictor on the same evaluation set as a reference. Conformal evaluation included marginal coverage, class-specific and worst-class coverage, worst-class undercoverage, absolute marginal-coverage error, mean set size, and empty-, singleton-, and doubleton-set fractions.

H2 paired each positive-budget target-fitted probability row with the same method's source-fitted budget-zero row within task, split, base model, model seed, adaptation seed, budget, and sample hash. Benefit-oriented differences were defined so that positive values indicated improvement. The prespecified material thresholds were an absolute log-loss reduction of at least 0.002, a relative log-loss reduction of at least 1%, and a Brier-skill gain of at least 0.005. Availability and algorithm-estimability rates used all attempts. Benefit and harm rates used only successful finite pairs; `not_available` and `not_estimable` were never interpreted as absence of harm.

H3 constructed complete budget curves only for the five tasks supporting all budgets through 500. Each cell retained absolute budget, target-pool fraction, and observed positive and negative label counts. Repeated acquisition effects were first collapsed within task. Cross-task summaries then used equal task weighting, with equal source-family weighting as a secondary view so that the two ACS tasks did not implicitly double the ACS family's contribution. Leave-one-task-out (LOTO) and leave-one-source-family-out (LOFO) analyses were recomputed from the paired H2 effects and interpreted as deletion-stability checks, not independent replications. College remained a separate 25–100 stress analysis. Target-pool fractions were task-specific descriptors of the tested absolute budgets; the protocol did not evaluate a shared grid of relative target-pool fractions.

H4 directly paired budget-zero results on `source_id_test` and `target_final_test`. Raw differences were target-final minus source-ID. The primary probability comparison used uncalibrated base-model scores; source-fitted calibrators were supplementary. H4 was descriptive: it assessed whether discrimination and reliability metrics could move differently under the evaluated domain shifts, without causal attribution or a claim that AUROC is independent of, or

irrelevant to, reliability.

H5 compared Mondrian with standard conformal prediction within each matched attempt. Metric differences were defined only when both methods completed successfully and the resulting contrast was finite. Within each task-model-budget cell, we took the median of the jointly successful paired differences, retaining successful Mondrian outputs with infinite thresholds in the primary conditional summary; we then averaged the 10 cell medians equally at each budget. A sensitivity analysis further restricted pairs to finite Mondrian thresholds and averaged the available cell medians. Mondrian estimability and finite-threshold rates were calculated over available attempts; all primary-set configurations at the five tested positive budgets were available. Positive worst-class-coverage differences represented higher observed Mondrian coverage, whereas positive mean-set-size differences represented larger prediction sets. Non-estimable attempts remained in the feasibility accounting and were not assigned a zero metric difference. Coverage, feasibility, and set-size effects were reported jointly; no single difference was used as an overall method ranking.

Descriptive summaries reported task/model-stratified effects, conditional event rates, eligibility rates, and deletion stability. These quantities represent descriptive heterogeneity rather than run-level confidence intervals. Task was the primary aggregation unit and source family the secondary unit. No run-level bootstrap, global run-level p -value, or row-level pseudo-replication was used. H1 retained predicted class-support probability, empirical class-support rate, empirical algorithm-estimability rate, and empirical finite-threshold rate as distinct quantities. Exact finite-population support under the observed pool composition was a retrospective validation only; deployment-style planning used the predeclared prevalence interval $[0.01, 0.99]$. Run-level binary-event MAE and RMSE were not interpreted as probability-versus-rate calibration error.

Completed simulations were used only as mechanism probes. The probability grid contained seven mechanisms—reference, mild and severe intercept shifts, slope shift, nonlinear shift, subgroup shift, and compressed scores—while the conformal grid contained reference, overconfidence, class-asymmetric, and rare-class mechanisms. The grids crossed target prevalences 0.01, 0.05, 0.20, and 0.50 with budgets 25–500 and 200 fixed repetitions per cell. Simulation rows were summarized separately from natural-shift rows and were not treated as external validation. In particular, the simulation-specific sigmoid support convention was not substituted for the natural-shift implementation's requirement that both outcome classes be represented.

All reported aggregates were deterministic transformations of the recorded machine-readable tables. For H2, model-specific task-equal summaries were averaged equally over logistic regression and CPU XGBoost; no run pooling was used. For H5, the primary conditional summary at each budget was an equal mean over the 10 primary task-model cells after taking the within-cell median over jointly successful paired metric differences. The finite-threshold sensitivity

summary averaged the available cell medians after additionally requiring a finite Mondrian threshold. Figures display those aggregates or the underlying evaluated task cells, and every panel has a CSV source-data file specifying the selected rows and transformation. This separation between experiment output and presentation output permits an audit of rounding, filtering, sign conventions, and denominators without reopening model execution.

7. Results

7.1 Experimental completion and eligibility

All 54 prespecified model configurations completed, covering six tasks, three split seeds, logistic regression, and the two CPU XGBoost model-seed strata. The experiment inventory contained no technical failure row. Probability results comprised 98,796 successful, 7,200 `not_available`, and 2,544 `not_estimable` rows; conformal results comprised 96,600 successful, 7,200 `not_available`, and 4,632 `not_estimable` rows. The threshold result family was a view of the same conformal attempts and was not added again when counting failures.

The combined method-level `not_estimable` records therefore contained 7,176 rows. Of the probability rows, 114 intercept-only MLE attempts and 114 sigmoid attempts lacked both classes, while 2,316 isotonic attempts lacked its required per-class support. The 4,632 conformal rows were Mondrian attempts with fewer than five labels in at least one class. All `not_available` positive-budget rows arose from prespecified task–budget combinations rather than technical termination; in particular, College budgets 250 and 500 remained explicit instead of being dropped.

The auxiliary H1 audit separated basic class support, method eligibility, and finite thresholds. Across the six tasks, empirical support for at least one label per class increased from 98.1% at budget 25 to 99.8% at 50 and 100% from 100 onward. Support for at least five labels per class was 69.2%, 89.4%, 98.6%, 100%, and 100% across budgets 25, 50, 100, 250, and 500. Method-specific eligibility remained a separately reported operational coordinate rather than an inferred performance outcome.

Finite Mondrian thresholds were more restrictive at small budgets. The corrected rate-level audit gave a predicted mean finite-threshold rate of 0.779157 and an empirical mean of 0.779643. Empirical rates across the five positive budgets were 0.3306, 0.7317, 0.9100, 0.9993, and 1.0000. These values describe the realized six-task design and do not convert known target composition into prospective deployment information. Detailed rate-level validation is reported in the supplement.

7.2 Complexity, budget, benefit and harm

Figure 2 and Table 3 show the primary five-task H2 results, averaged equally over the two prespecified base-model strata after task-level aggregation. Positive change denotes lower target-final log loss relative to the same update fitted on the source probability-calibration role at budget zero. Material harm denotes a paired log-loss change of at most -0.002 and is calculated only among successful finite pairs. Eligibility is shown separately, so a method that could not be fitted was not counted as harmless.

Complexity–budget response for the four probability updates. Panel A shows the task-equal mean, task–model interquartile range, and minimum-to-maximum range on a symmetric logarithmic scale; these descriptive spreads are not confidence intervals. The dashed line marks the $+0.002$ material-benefit threshold. Panel B shows conditional material-harm rates.

Intercept-only MLE moved from a near-zero mean benefit of -0.000103 at budget 25 to $+0.004040$ at 50, $+0.006711$ at 100, and $+0.008803$ at 500. Its material-harm rate nevertheless remained 43.1% at budget 25 and 30.4% at 50 before falling to 15.0%, 4.2%, and 1.0% at budgets 100, 250, and 500. Thus, the earlier average recovery of this low-dimensional update did not make the smallest budget risk-free.

Jeffreys smoothing was estimable in 100% of primary-set attempts at every positive budget. Its mean benefit changed from $+0.000342$ at budget 25 to $+0.008680$ at 500, while its harm rate changed from 43.6% to 1.0%. The corresponding curves closely tracked unsmoothed intercept-only MLE. The recorded evidence therefore supports a low-budget availability advantage, but not a clear performance advantage over the unsmoothed intercept update.

Sigmoid calibration's mean log-loss benefit was -0.025451 at budget 25 and -0.006135 at 50. The mean became positive at budget 100 ($+0.002534$), one tested budget step later than intercept MLE, but 33.5% of successful finite pairs still met the material-harm definition. At budget 250 the mean benefit was $+0.006874$ and harm fell to 8.9%; at 500 these values were $+0.008486$ and 1.7%. The average sign at budget 100 therefore did not by itself mark a low-risk point.

Isotonic calibration became increasingly estimable—from 64.5% at budget 25 to 87.3% at 50, 98.3% at 100, and 100% at 250 and 500—but its primary-set mean log-loss benefit remained negative: -0.4218 , -0.2595 , -0.1365 , -0.0477 , and -0.01722 . Its conditional material-harm rate declined from 99.7% to 73.2% across the same grid. Persistent harm at the upper tested budget was therefore not explained by ineligibility alone. This statement is confined to the evaluated natural-shift grid and does not characterize isotonic behavior in other settings.

Table 3. Common-five probability-update effects, material-benefit heterogeneity, and deletion stability. Mean change is task-equal and then equal over the two model strata.

Method	Budget	Mean Δ log loss	Material harm	Estimable	Material- positive task-model cells / 10	Worst task-model median	LOTO expected direction retained / 10
Intercept MLE	25	-0.0001	43.1%	97.7%	4	-0.0120	5
Intercept MLE	50	+0.0040	30.4%	99.8%	5	-0.0050	9
Intercept MLE	100	+0.0067	15.0%	100.0%	6	-0.0026	10
Intercept MLE	250	+0.0082	4.2%	100.0%	6	-0.0010	10
Intercept MLE	500	+0.0088	1.0%	100.0%	7	-0.0006	10
Jeffreys intercept	25	+0.0003	43.6%	100.0%	5	-0.0116	6
Jeffreys intercept	50	+0.0044	29.4%	100.0%	5	-0.0050	10
Jeffreys intercept	100	+0.0067	14.3%	100.0%	6	-0.0026	10
Jeffreys intercept	250	+0.0081	4.3%	100.0%	7	-0.0010	10
Jeffreys intercept	500	+0.0087	1.0%	100.0%	7	-0.0006	10
Sigmoid	25	-0.0255	74.0%	97.7%	0	-0.0400	10
Sigmoid	50	-0.0061	57.3%	99.8%	2	-0.0149	10
Sigmoid	100	+0.0025	33.5%	100.0%	4	-0.0052	8
Sigmoid	250	+0.0069	8.9%	100.0%	5	-0.0012	10
Sigmoid	500	+0.0085	1.7%	100.0%	6	-0.0004	10
Isotonic	25	-0.4218	99.7%	64.5%	0	-0.5634	10
Isotonic	50	-0.2595	98.3%	87.3%	0	-0.3456	10
Isotonic	100	-0.1365	95.1%	98.3%	0	-0.1733	10
Isotonic	250	-0.0477	86.5%	100.0%	0	-0.0702	10
Isotonic	500	-0.0172	73.2%	100.0%	2	-0.0342	10

7.3 No common tested absolute budget within the evaluated grid

No evaluated method–budget combination achieved the prespecified $+0.002$ log-loss improvement in all 10 common task–model cells. Figure 3 makes the source of this result visible: task-specific median effects differed in magnitude and, at several budgets, in sign within both logistic regression and CPU XGBoost. Intercept updates generally entered positive regions earlier than sigmoid, but individual tasks did not share one boundary. Isotonic remained negative in most cells even when its task-specific curve improved with budget.

Task-, model-, method-, and budget-specific median log-loss benefit across the primary five-task set. Each cell is a task–model summary; College is excluded. The symmetric logarithmic color scale retains small changes around zero while showing large isotonic harms; open circles mark absolute effects below 0.002.

Absolute label counts also represented different fractions of the target adaptation pools. At budget 100, the split-level H3 curves gave fractions of 0.087% for BRFSS Diabetes, 0.296% for ACS Income, 0.663% for Diabetes Readmission, 1.136% for ACS Food Stamps, and 1.970% for NHANES Lead—a 22.6-fold range between the smallest and largest common-task fractions. These fractions describe how the tested absolute budgets mapped to each task; the study did not evaluate a common relative-budget grid and therefore does not identify a universal target-pool-fraction threshold.

The principal directions were stable to the prespecified deletion checks: 169 of 170 LOTO checks and all 136 LOFO checks retained their expected direction. The single LOTO reversal had magnitude approximately -0.000028 . These analyses show that the observed pattern was not driven solely by one task or one source family, but they are deletion-stability descriptions within the observed sample, not additional population replications.

College Scorecard remained separate. Its registry adaptation capacity was 107, while the executed split-level H3 curve placed budget 100 at approximately 96.2% of the available adaptation pool. Budgets 250 and 500 were `not_available`. The lower-budget effects also showed a strong logistic-regression versus XGBoost interaction, reinforcing the decision not to pool this small-domain stress task with the complete primary-set curves.

7.4 Class-conditional reliability and set-size cost

Among jointly successful standard–Mondrian pairs across the primary five-task set, two base models, five positive budgets, and the primary uncalibrated score source, the median difference in worst-class coverage was positive in each of the 50 task–model–budget cells. Thirty-eight cell medians were at least 0.10, and the direction agreed between the two model strata in all 25 task–budget combinations. These metric comparisons are conditional on both methods producing successful results; estimability and finite-threshold rates are reported separately

over available attempts.

Paired Mondrian-minus-standard results. Panels A and B compare the joint-success conditional summary, which retains infinite Mondrian thresholds, with a sensitivity analysis restricted to jointly successful pairs with finite Mondrian thresholds. Panel C reports Mondrian estimability and finite-threshold rates across available attempts.

The joint-success conditional mean worst-class-coverage difference rose from +0.1285 at budget 25 to +0.2876 at 50, +0.3420 at 100, +0.3505 at 250, and +0.3541 at 500. Mean set size in the same conditional summary simultaneously increased by +0.3280, +0.3469, +0.2843, +0.2069, and +0.1975. The associated doubleton-fraction changes were +0.3278, +0.3465, +0.2776, +0.2030, and +0.1923, showing that most of the binary set expansion came from full two-label outputs rather than a change between empty and singleton sets.

Feasibility changed sharply with budget. Across available attempts, Mondrian estimability was 64.5%, 87.3%, 98.3%, 100%, and 100% across budgets 25–500; finite-threshold rates were only 30.1% and 68.2% at 25 and 50, then 89.2%, 99.9%, and 100%. In the finite-threshold-only successful subset at budget 25, the mean worst-class gain was approximately +0.0275 and the mean set-size change approximately +0.0013. This descriptive subset represented only four tasks and eight task–model cells because NHANES had no finite row at that budget. The larger joint-success conditional budget-25 averages were therefore strongly influenced by infinite thresholds and full-label sets. This low-budget distinction is expanded in the supplement.

Marginal coverage and absolute marginal-coverage error did not improve uniformly. Mean raw Mondrian-minus-Standard marginal-coverage changes were -0.0085 , $+0.0083$, $+0.0084$, $+0.0038$, and $+0.0029$ over the five budgets, while absolute-error differences changed sign. The H5 result is thus a class-conditional coverage–feasibility–set-cost trade-off, not an overall ordering of Standard and Mondrian.

Table 4. Paired Mondrian-minus-standard conditional metric differences, feasibility, and finite-threshold sensitivity.

Budget	Joint-success Δ worst-class coverage	Joint-success, finite-threshold Δ worst-class coverage	Joint-success Δ mean set size	Joint-success, finite-threshold Δ mean set size	Finite task-model cells	Mondrian estimable (available attempts)	Finite threshold (available attempts)
25	+0.1285	+0.0275	+0.3280	+0.0013	8/10	64.5%	30.1%
50	+0.2876	+0.2044	+0.3469	+0.1984	10/10	87.3%	68.2%
100	+0.3420	+0.3167	+0.2843	+0.2227	10/10	98.3%	89.2%
250	+0.3505	+0.3505	+0.2069	+0.2068	10/10	100.0%	99.9%
500	+0.3541	+0.3541	+0.1975	+0.1975	10/10	100.0%	100.0%

Metric-difference columns condition on joint success; the primary joint-success columns retain successful Mondrian outputs with infinite thresholds. Feasibility columns use available-attempt denominators, and non-estimable attempts are not assigned zero differences.

7.5 Auxiliary discrimination–reliability diagnostic

The primary H4 diagnostic contained 54 strictly paired, budget-zero source-ID and target-final comparisons using uncalibrated base-model probabilities. Among the 17 pairs with strict $|\Delta\text{AUROC}| < 0.01$, Brier score and log loss both worsened in 17 of 17; their median raw increases were +0.0271 and +0.0717. At the strict 0.02 threshold, both scores worsened in all 32 of 32 pairs, with median increases +0.0182 and +0.0467. These are task–split–model–design pairs, not independent task replications; the 17-pair threshold set is nested within the 32-pair set. Thus, small AUROC changes did not obviate direct evaluation of Brier score and log loss in these comparisons.

Source-to-target changes for the 54 primary uncalibrated H4 pairs. Raw horizontal differences are target-domain test minus source-domain test AUROC; positive vertical differences are worse Brier score or log loss. Dashed lines mark absolute AUROC changes of 0.01; star markers identify the separate College stress task, and the horizontal axis uses a symmetric-log transform.

Across all 54 design pairs, AUROC changes were strongly negatively associated with Brier-score and log-loss changes: Pearson correlations were -0.908 and -0.895 ; the corresponding Spearman correlations were -0.762 and -0.772 . Because multiple pairs arose from the same tasks, these descriptive correlations are not population-level estimates. AUPRC and calibration-coefficient diagnostics also did not all move in one direction. AUROC remains a discrimination measure; the bounded result is that it did not replace direct probability-reliability evaluation.

7.6 Mechanism-probe simulations

The completed probability simulations qualitatively reproduced the small-budget sensitivity of more complex updates, and the conformal simulations reproduced a class-asymmetry mechanism in which classwise coverage could require larger sets. They did not reproduce all natural-shift behavior. Simulated isotonic mean log-loss benefit was approximately -0.410 and -0.172 at budgets 25 and 50, then became positive at about +0.058, +0.197, and +0.240 at budgets 100, 250, and 500. In contrast, the primary-set natural-shift mean remained negative through 500.

The archived analysis review placed sigmoid's simulated reduction in harm near budget 100. In the natural-shift analysis, however, 33.5% of its successful finite pairs at budget 100 still met the material-harm criterion, and the rate fell below 10% only at budget 250 (8.9%). The simulation grid therefore supports a broad

complexity–sample-size mechanism while disagreeing on the budget location and high-budget isotonic pattern. Conformal finite-threshold rates in the simulation summary were conditional on estimable successes, so they were not directly subtracted from the available-attempt natural-shift rates. The simulations did not contain the source-domain/target-domain test pairs required for H4.

8. Discussion

The central empirical result is that update complexity and target-label budget acted jointly, but the ordering of mean benefits across tested budgets did not yield one budget that satisfied the prespecified improvement criterion across task–model–method cells. Intercept updates were the lower-dimensional anchors, and their mean benefit became positive at a lower tested budget; sigmoid generally required more labels to reduce harm; and isotonic retained substantial log-loss harm within the tested natural-shift range. Yet even the intercept methods had material harm in more than two fifths of successful budget-25 pairs. A mean positive effect is therefore insufficient for a reliability decision when the distribution of paired harms remains wide.

This pattern is consistent with established work on calibration sample size and staged prediction-model updating (Niculescu-Mizil and Caruana 2005; Steyerberg et al. 2004; Vergouwe et al. 2017; Ojeda et al. 2023; Sun et al. 2023). The contribution here is not the qualitative mechanism. It is the joint measurement of benefit, material harm, availability, target-pool fraction, and deletion stability under one label-blind random protocol spanning multiple natural tabular shifts and two base predictors. This protocol-level view exposes a decision problem that a method-average score can hide: whether a proposed update can be fitted, how frequently it produces a material loss, and whether the same nominal budget has comparable meaning across targets.

H3 sharpens that point. A budget of 100 labels ranged from a very small fraction of a large BRFSS adaptation pool to almost the entire executed College pool. Relative budget alone also cannot encode minority-class support, score structure, base-model calibration, or update dimension. Because a common relative-budget grid was not tested, the observed pool fractions should be read as task-specific descriptors rather than as evidence for or against a universal fraction threshold. The appropriate interpretation is conditional: for the evaluated tasks, the evidence can identify regions with lower observed risk, but it cannot turn budget 250—or any other tested value—into a general operating rule. The LOTO and LOFO checks indicate that no single task or source family alone determined the overall direction; however, these remain descriptive sensitivity analyses within the observed task set.

The Jeffreys comparison illustrates why availability and performance should be separated. Smoothing removed the requirement that both outcome classes be represented and produced complete low-budget estimability, but its benefit and

harm curves stayed close to unsmoothed intercept MLE. For a user who values a finite update in a single-class sample, that availability is operationally relevant. It is not evidence that smoothing improves predictive performance in general. Similarly, isotonic's rising estimability did not make its fitted updates beneficial on average in the evaluated natural tasks. Eligibility is a prerequisite for a performance comparison, not a performance result.

The H5 findings add a separate class-conditional decision layer. Among jointly successful pairs, Mondrian increased observed worst-class coverage in every primary task-model-budget cell, but low-budget thresholds were often infinite and prediction sets expanded. In a binary problem, that expansion was mainly an increase in doubletons, which express uncertainty by returning both labels. Whether this is acceptable depends on the decision context: an application that prioritizes the class with lower observed coverage may accept a larger set, whereas an application requiring actionable singleton outputs may not. Recent class-conditional conformal research already establishes this coverage-efficiency tension and develops other ways to navigate it (Ding et al. 2023, 2026; Shi et al. 2024; Liu et al. 2026; Papaioannou et al. 2026; Althani 2026). Our result contributes the paired budget response under the same natural-tabular adaptation samples, not a claim that the tension itself was previously unknown.

Retaining unsuccessful and non-estimable attempts changes the interpretation of both probability and conformal results. A method with attractive conditional performance but low eligibility cannot be compared directly with a method that produces a result in every attempt unless the availability difference is visible. The explicit `not_available`, `not_estimable`, `infinite-threshold`, and `failed` states prevent an apparently clean curve from being created by deleting difficult samples. They also separate different remedies: acquiring more total labels addresses some support failures, smoothing changes an intercept fit's eligibility rule, and finite classwise conformal thresholds require adequate support in each outcome class.

The H4 diagnostic reinforces the need for a multi-metric audit without devaluing discrimination. AUROC answers a ranking question, whereas Brier score and log loss assess probability quality. The descriptive associations between their source-to-target changes were strong, but the near-zero-AUROC subsets demonstrate that a small ranking change did not rule out a meaningful proper-score deterioration. This distinction is established in the calibration and model-shift literature (Davis et al. 2017; Levy et al. 2022; Guo et al. 2023; Ovadia et al. 2019; Van Calster et al. 2019); the present counts are auxiliary cross-task evidence under the prespecified protocol.

Simulation provides a final caution against turning a plausible mechanism into a complete account of the natural results. The simulated complexity ordering and conformal set expansion were directionally informative, but simulated isotonic calibration improved at high budgets while the natural-shift aggregate did not. Natural shifts can combine prevalence, covariate, conditional, and score-geometry changes that are simplified in a mechanistic grid. The disagreement

is scientifically useful because it bounds the interpretation: the simulation can illustrate why small samples may destabilize a flexible update, but it cannot override the empirical behavior of the evaluated natural tasks.

Taken together, the results suggest an audit sequence for similar bounded studies. Investigators can define entity-disjoint source and target roles, acquire labels without outcome-aware selection, record class support and method eligibility, compare each update with its own source-fitted reference, and examine benefit and material harm before selecting a budget. If class-conditional prediction sets are considered, finite thresholds and set composition should accompany coverage. This sequence is a reporting and decision discipline derived from the evaluated protocol, not a claim that the tested methods or budgets transfer unchanged to another application.

The results also argue for separating a budget recommendation into at least three questions. The sampling question asks how much of the target pool is labeled and which class counts are likely under a label-blind draw. The estimation question asks whether the chosen updater or classwise threshold can be fitted under that support. The performance question asks whether successful fits improve a proper score or coverage objective often enough, and at an acceptable cost. A method can pass one question and fail another: Jeffreys smoothing can be available without outperforming its unsmoothed analogue, isotonic can become estimable while remaining harmful on average, and Mondrian can increase worst-class coverage while returning more doubletons. Reporting these questions separately makes task-specific trade-offs visible and reduces the temptation to treat a positive mean or a nominal label count as a complete decision.

9. Limitations

The empirical sample is limited to six public binary tabular tasks from five source families. The primary pooled results use only five tasks from four families because College lacks the two upper budgets. Two base predictors—logistic regression and one CPU XGBoost configuration family—cannot represent the behavior of other boosting systems, random forests, support-vector machines, neural tabular models, foundation models, ensembles, or models trained with different tuning objectives. The study does not cover multiclass outcomes, regression targets, structured outputs, or repeated-event prediction.

The update set is also narrow. Probability analyses are limited to the pre-specified uncalibrated, intercept-only MLE, Jeffreys-smoothed intercept, sigmoid, and isotonic implementations. Conformal analyses compare Standard and outcome-class Mondrian methods with two fixed score sources and nominal coverage 0.90. The study does not evaluate beta or temperature calibration, arbitrary nonparametric updates, full model refitting, representation adaptation, weighted or online conformal methods, or hierarchical classwise fallbacks. H6 was outside the final scope, so the evidence says nothing about outcome-aware,

active, or alternative label-acquisition policies.

Budgets are restricted to 25, 50, 100, 250, and 500 label-blind random target labels, with budget zero as a source-fitted comparison reference. The discrete grid does not identify an exact transition between tested values, and there is no evidence beyond 500. Absolute counts and target-pool fractions do not fully determine realized per-class support. College is a separate stress case with only the lower budgets, and its high sampling fraction should not be extrapolated to larger tasks.

The source–target domains are public observational partitions defined by geography, survey group, poverty, admission source, or institutional control. They represent natural shifts but are not randomized interventions. The analysis cannot identify why a metric changed, isolate a causal shift component, establish class or group fairness beyond the reported outcome-class coverage, or certify safe deployment. Results are also bounded to the countries, periods, inclusion rules, feature definitions, class prevalences, and public-data measurement processes represented by these six tasks.

Repeated splits, model seeds, and adaptation samples improve description of the fixed design but do not create independent task replications. XGBoost seed strata are model-design repetitions, and H1 model rows duplicate the same label-sampling design calculation. Task-equal and family-equal summaries, together with LOTO and LOFO analyses, describe sensitivity within this sample. They do not establish a population distribution over tabular shifts. Harm rates are conditional on successful finite pairs and must be read alongside the separate eligibility rates.

The conformal comparison is empirical under the prespecified source–target construction. Natural shift can violate exchangeability assumptions that underlie standard finite-sample conformal statements; the paper reports realized target coverage rather than asserting a distribution-free target-domain property. At low budgets, infinite Mondrian thresholds can produce full binary sets, so higher worst-class coverage need not mean more informative prediction. Marginal coverage and absolute coverage error also did not improve consistently.

The H4 analysis is a diagnostic association. It does not show that AUROC change causes proper-score change, that the metrics are independent, or that discrimination lacks value. The simulations are mechanism probes rather than held-out natural tasks. Their restricted data-generating mechanisms do not reproduce every feature of real domain shift, and the high-budget isotonic disagreement is a material limitation rather than a result to average away. The simulation grid also lacks the source-ID/target-final structure needed to examine H4.

Finally, literature positioning is bounded to the completed search and verification record. No complete duplicate of the joint protocol was identified in that record, but inaccessible or later work may alter this assessment. No claim in this article relies on literature that was available only as unverified metadata.

10. Conclusion

Within the primary five-task set, two tested base models, four probability updates, and label-blind random target-label budgets of 25–500, calibrator complexity and budget jointly shaped mean benefit and the observed material-harm rate among successful finite pairs. The mean benefit of intercept updating became positive at a lower tested budget than that of sigmoid, while isotonic retained negative mean log-loss benefit through the upper tested budget. No tested absolute budget met the prespecified improvement condition across the evaluated task–model–calibrator grid. Because the design did not test a common relative-budget grid, it does not identify a universal fraction threshold.

The class-conditional analysis reached a complementary conclusion. Among jointly successful pairs, the median Mondrian-minus-standard worst-class-coverage difference was positive in all 50 primary task–model–budget cells, but estimability, finite-threshold rates, and prediction-set expansion varied strongly with budget. The observed coverage differences were therefore inseparable from feasibility and binary set cost. The bounded practical implication is to evaluate method, task, model, class support, and budget together, retain unsuccessful attempts in the record, and report probability and set-valued reliability on their own metrics.

Data and Code Availability

The study used public secondary datasets governed by the licenses and terms recorded in the repository's data-license registry; official source and access details are listed in Supplementary Table S8. The public release contains manuscript artifact sources, figure- and table-level machine-readable source data, analysis and artifact-generation code, environment specifications, checksums, and a documented workflow for one representative end-to-end ACS model shard. An optional workflow re-executes the 54 prespecified model shards from prepared, audited Parquet inputs; it does not reconstruct every source dataset or regenerate the complete intermediate analysis tree. The release does not redistribute raw third-party datasets. Data-acquisition instructions identify official sources and applicable provider terms. Author-created software is licensed under the MIT License. The public repository URL, immutable archival DOI, and release tag will be inserted after the release is deposited and before submission.

Funding

The author received no specific funding for this work.

Competing Interests

The author declares that there are no competing interests.

Author Contributions / CRediT

Dinghao Chen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, and Writing - review & editing. The author accepts full responsibility for the work.

Acknowledgments

The author thanks the College of Big Data, Baoshan University, for providing access to the computing facilities used in this study.

Declaration of Generative AI-Assisted Work

OpenAI Codex (GPT-5; accessed 11-14 July 2026) was used solely for manuscript formatting, language refinement, file-consistency checks, and pre-submission review to improve clarity, consistency, and compliance with submission requirements. It was not used to formulate the research question, design the study, conduct experiments, analyze data, interpret results, formulate scientific claims, or draw conclusions; these activities were completed independently by the author. The author reviewed all AI-assisted changes, confirmed the originality and accuracy of the manuscript, verified all references, checked the tool's terms of use and suitability for publication, retained the original and revised versions, and accepts full responsibility for the integrity of the content. Generative AI was not used to create or modify research data or the scientific features of any figure.

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